

Agents Exercise

Team Members: _____

Problem chosen: _____

Pick a problem from below or invent your own:

1) “Assignment Autograder Helper”

User goal: “Given my lab notebook + code, tell me what’s failing and how to fix it.”

2) “Resume Optimizer with Claim Verification”

User goal: “Tailor my resume to this job posting and ensure my bullets are defensible.”

3) “Research Digest Builder”

User goal: “Given 5 URLs, produce a 1-page brief with themes, pros/cons, and citations.”

4) Choose your own!

Requirements

- Use at least 3 tools (more is fine).
- Show an iterative agent loop (2–4 iterations) using a Thought → Action (tool,args) → Observation format.
- Name the fields passed between modules (don’t pass “everything”).
- Include a stopping rule (e.g., max_iters, success condition, or ask-user condition).
- List two likely failure modes and one test/metric for each.

What to submit

- A. End-to-end task definition (success criteria + inputs available).
- B. Tool list with contracts (inputs/outputs).
- C. Agent loop diagram (boxes + arrows).
- D. Example trajectory (2–4 iterations).

Completed example (use this as a template)

Example problem: Meeting Scheduler

A) Task definition

- Success criteria:
 - Propose 2–3 candidate 30-minute slots within working hours (9am–5pm).
 - Respect each person’s existing calendar busy blocks.
 - Draft a short confirmation message with the chosen slot + alternates.
- Inputs available:

- meeting_request: {attendees: list[str], duration_min: int, date_window: {start, end}, working_hours: {start_hour, end_hour}}
- calendar data (busy blocks per attendee) via tools

B) Tools (contracts)

- **calendar_lookup**(person_id: str, window: {start: datetime, end: datetime})
 - → busy_blocks: list[{start: datetime, end: datetime}]
- **find_common_free**(busy_by_person: dict[str, list[{start, end}]], duration_min: int, working_hours: {start_hour: int, end_hour: int})
 - → candidate_slots: list[{start: datetime, end: datetime}]
- **draft_message**(to: list[str], subject: str, body_fields: {chosen_slot, alternates})
 - → message_text: str

C) Loop sketch (ReAct-style)

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[User Request]
↓
[Parse constraints / state]
↓
[Decide next tool] → [Call tool] → [Observation] → [Update state]
↓
[Stop? (slots found OR max_iters OR ask user)]
↓
[Draft message] → [Final answer]
  
```

D) Example trajectory (3 iterations)

- **Thought:** Need busy blocks for all attendees next week.
- **Action:** calendar_lookup(person_id='Alice', window=next_week)
- **Observation:** Alice busy: Mon 10–11, Tue 1–3
- **Action:** calendar_lookup(person_id='Bob', window=next_week)
- **Observation:** Bob busy: Mon 9–10:30, Wed 11–12
- **Action:** calendar_lookup(person_id='Casey', window=next_week)
- **Observation:** Casey busy: Tue 2–5, Fri 1–2
- **Thought:** Compute 30-min common free slots in 9–5.
- **Action:** find_common_free(..., duration_min=30, working_hours={9,17})
- **Observation:** candidates: Mon 11:00–11:30, Wed 10:00–10:30, Fri 10:30–11:00
- **Thought:** Choose earliest slot and draft message.
- **Action:** draft_message(to=[...], subject='Meeting next week', body_fields={chosen, alternates})
- **Observation:** message_text returned; ready to send.

Choose A, B, C, or D. Fill in the sections below.

Success criteria (2–3 bullets):

- _____
- _____
- _____

- _____
- _____
- _____

Tool name

Outputs (fields)

Draw your loop here (include stop condition and what triggers tool calls):

Write 2–4 iterations using Thought → Action(tool,args) → Observation:

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